

HEISENBERG GROUP AND REALITY

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1

Let H be the complex Heisenberg group, an extension of $\mathbb{C}^2$ by $\mathbb{C}^*$. It has coordinates (x, y, λ) , with group law

$$(x, y, \lambda)(x', y', \lambda') = (x + x', y + y', \lambda\lambda' \exp(\pi i(xy' - x'y))). \quad (1.1)$$

The involution

$$T: (x, y, \lambda) \mapsto (-\bar{x}, -\bar{y}, \bar{\lambda}) \quad (1.2)$$

is an automorphism of H .

The coordinate system $x, y, \lambda: H \xrightarrow{\sim} \mathbb{C}^2 \times \mathbb{C}^*$ displays the section $\mathbb{C}^2 \times \{1\}$ of the projection of H onto $\mathbb{C}^2$. On this section one has $T(g) = g^{-1}$. Each line $D \subset \mathbb{C}^2$ has, as its lift by this section, a one-parameter subgroup of H , which we again denote by D . The commutator $(u, v) = uvu^{-1}v^{-1}$ of two elements $u = (u_1, u_2)$ and $v = (v_1, v_2)$ of $\mathbb{C}^2$ is $\exp(2\pi i \langle u, v \rangle) \in \mathbb{C}^*$, where

$$\langle u, v \rangle = u_1 v_2 - u_2 v_1. \quad (1.3)$$

Let $H_{\mathbb{R}}$ be the real form defined by $x, y \in \mathbb{R}$ and $|\lambda| = 1$. The group $\mathrm{SL}(2, \mathbb{C})$ acts on H , and $\mathrm{SL}(2, \mathbb{R})$ acts on $H_{\mathbb{R}}$, which is identified set-theoretically with $\mathbb{R}^2 \times \mathrm{U}^1$.

It is known, by the Stone–von Neumann theorem [2], that $H_{\mathbb{R}}$ has a unique irreducible unitary representation V in which $\lambda \in \mathrm{U}^1$ acts by multiplication by λ . In the Schrödinger model of V , one has $V = L^2(\mathbb{R})$ and the action of $H_{\mathbb{R}}$ is

$$\begin{aligned} (X, 0, 1): f(z) &\mapsto f(z + X), \\ (0, y, 1): f(z) &\mapsto e^{2\pi i y z} f(z). \end{aligned} \quad (1.4)$$

From the uniqueness of V one obtains a projective unitary action of $\mathrm{SL}(2, \mathbb{R})$ on V , compatible with the action of $\mathrm{SL}(2, \mathbb{R})$ on $H_{\mathbb{R}}$ (A. Weil [3], D. Shale [1]).

For the action of $H_{\mathbb{R}}$ on $V = L^2(\mathbb{R})$, the C^∞ vectors, i.e. those $v \in V$ for which $h \mapsto hv$ is C^∞ in $h \in H_{\mathbb{R}}$, form the Schwartz space $\mathcal{S}$ of C^∞ rapidly decreasing functions on $\mathbb{R}$. The holomorphic vectors, i.e. those $v \in V$ for which $h \mapsto hv$ extends to a holomorphic map from H to V , are the L^2 functions on $\mathbb{R}$ which extend to holomorphic functions on $\mathbb{C}$ such that, in every horizontal strip $|\mathrm{Im} z| < A_1$, one has for every $A_2 \in \mathbb{R}$

$$|f(z)| = O(\exp(-A_2|z|)).$$

The complex group H acts on the space V^{hol} of holomorphic vectors by the formulae (1.4).

For the holomorphic action of H on V^{hol} one does not have the same uniqueness as for the unitary action of $H_{\mathbb{R}}$ on V . If one did, the projective action of $\text{SL}(2, \mathbb{R})$ on V would extend to a projective action of $\text{SL}(2, \mathbb{C})$ on V^{hol} ; this action would lift to a genuine action of $\text{SL}(2, \mathbb{C})$, since $\text{SL}(2, \mathbb{C})$ is simply connected; and the projective action of $\text{SL}(2, \mathbb{R})$ would lift to a genuine action. This is not the case: only the double cover of $\text{SL}(2, \mathbb{R})$ acts. Our purpose is to make explicit what happens.

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Let $S = \mathbb{P}^1(\mathbb{C})$ be the space of lines in $\mathbb{C}^2$. For $D \in S$, lifted into H as in no. 1, let W_D be the space of holomorphic functions on H satisfying

$$\varphi(\lambda d \cdot h) = \lambda \varphi(h)$$

for $\lambda \in \mathbb{C}^*$ and $d \in D$, endowed with the action of H given by

$$g: \varphi(h) \mapsto \varphi(hg).$$

For $E \not\supset D$, restriction to E is a bijection

$$r_E: W_D \xrightarrow{\sim} \mathcal{O}(E), \tag{2.1}$$

where $\mathcal{O}(E)$ is the space of holomorphic functions on E .

If R is a holomorphic representation of H , on which each $\lambda \in \mathbb{C}^*$ acts by λ , and if ω is a D -invariant linear form on R , then

$$\omega \mapsto \omega(hw) \tag{2.2}$$

is a morphism from R to W_D .

3. Example

Let V be, as in nos. 1 and 2, the irreducible unitary representation of $H_{\mathbb{R}}$, let $\mathcal{S} \subset V$ be the space of its C^∞ vectors, and let $\mathcal{S}' \supset V$ be the dual of the space of C^∞ vectors of the dual representation; in the Schrödinger model this is the space of tempered distributions.

Let $D_t \in S$ be the line generated by $(-t, 1)$. If $\text{Im } t > 0$, the linear form, written in the Schrödinger model,

$$\omega_t: f \mapsto \int f(x) \exp(-\pi i x^2/t) dx$$

is defined on V , and even on $\mathcal{S}$. The form ω_t is a holomorphic vector in the dual of V , fixed by D_t . In terms of the infinitesimal generator $-t\partial_x + 2\pi i x$ of D_t , invariance is expressed by

$$\int [(-t\partial_x + 2\pi i x)f(x)] \exp(-\pi i x^2/t) dx = \int f(x)(t\partial_x + 2\pi i x) \exp(-\pi i x^2/t) dx = 0.$$

By (2.2), ω_t defines

$$Q_t: V^{\text{hol}} \rightarrow W_{D_t},$$

which extends to V and even to $\mathcal{S}'$.

Let E be the x -axis. The map $r_E \circ Q_t: V \rightarrow \mathcal{O}(E)$ sends $f \in V$ to

$$f(x, t) = r_E \circ Q_t(f) \quad (\text{Im } t > 0),$$

a solution of the heat equation

$$\partial_t f = \frac{i}{4\pi} \partial_x^2 f, \quad \text{i.e.} \quad \left(\partial_t - \frac{i}{4\pi} \partial_x^2 \right) f(x, t) = 0.$$

For real t , the form ω_t remains defined on $\mathcal{S} \subset V$. For $t = 0$ it is $f \mapsto f(0)$, and for $t = \infty$ it is $\int f(x) dx$. The homomorphism Q_t remains defined on V^{hol} . For $t = 0$, $r_E \circ Q_0$ is the identity in the Schrödinger model.

Assume $\text{Im}(t) > 0$. Let $\bar{D}_t$ be the complex conjugate line of D_t . In the affine line $S - \{\bar{D}_t\}$, the “half-plane” $\{D_u \mid \text{Im } u > 0\}$ is a disk centered at D_t . The map $r_{\bar{D}_t} \circ Q_t: V \rightarrow W_{D_t} \rightarrow \mathcal{O}(D_t)$ gives the holomorphic model of V : the action of H is given by

$$\begin{aligned} d \in D_t: f(x) &\mapsto f(x + d), \\ e \in D_t: f(x) &\mapsto \exp(-2\pi i \langle e, x \rangle) f(x), \end{aligned} \quad (3.1)$$

and the Hilbert structure for which (f, f) is the integral of

$$f(d) \overline{f(d')} \exp(2\pi i \langle d, d' \rangle)$$

is invariant under $H_{\mathbb{R}}$ and identifies V , via $r_{\bar{D}_t} \circ Q_t$, with the space of functions f of finite norm. For $d = (-tz, z)$, the factor $\exp(2\pi i \langle d, \bar{d} \rangle)$, the commutator of d and $\bar{d}$, is

$$\exp(2\pi i \langle d, \bar{d} \rangle) = \exp(2\pi i (t - \bar{t}) |z|^2).$$

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Fix E . The isomorphisms r_E in (2.1) allow one to regard the W_D with $D \not\supset E$ as forming a family of representations of H acting on the fixed space $\mathcal{O}(E)$, the action of H depending holomorphically on D . We spell out the dependence on E of r_E . Choose coordinates as in no. 1, with E the x -axis. Let E' be generated by $(1, u)$ and let D be generated by $(-t, 1)$. We take x as coordinate on both E and E' . Then

$$(x', ux', 1) = (-tux', ux', 1)(x, 0, 1)(0, 0, \lambda)$$

for $(1 + tu)x' = x$ and $\lambda = \exp(\pi i u x x')$. Thus $r_{E'} \circ r_E^{-1}$ sends $f_E \in \mathcal{O}(E)$ to $f_{E'} \in \mathcal{O}(E')$ with $f_{E'}(x') = \lambda f_E(x)$:

$$f_{E'} \left(\frac{x}{1 + tu} \right) = \exp \left(\frac{\pi t i}{1 + tu} x^2 \right) f_E(x). \quad (4.1)$$

This formula is holomorphic in t : the W_D form a holomorphic vector bundle $\mathcal{V}$ of infinite dimension on S , on which H acts.

To understand (4.1), one may note that the point of E with coordinate x has the same image in $\mathbb{C}^2/D$ as the point of E' with coordinate $x/(1 + tu)$.

From (4.1) one deduces that the following growth condition on $v \in W_D$ is independent of the choices of $E \not\supset D$ and of the coordinate z on E :

$$\text{for } f = r_E(v), \quad \exists A \quad |f(z)| < O(\exp(A|z|^2)). \quad (4.2)$$

Let $W_D^? \subset W_D$ be defined by this condition. The $W_D^?$ form a holomorphic subbundle $\mathcal{V}^?$ in holomorphic representations of H inside $\mathcal{V}$.

5. Proposition

Let $T > 0$ and consider the following conditions on an entire function f :

- (i)_T There is a holomorphic function $g(z, t)$, for $z \in \mathbb{C}$ and $|t| < T$, satisfying the heat equation $(\partial_t - \partial_z^2)g = 0$ and the initial condition $g(z, 0) = f(z)$;
- (ii)_T $|f(z)| < O(\exp(|z|^2/4T))$;
- (iii)_T $\iint |f(z)\overline{f(\bar{z})}| \exp(-|z|^2/2T) |dz \wedge d\bar{z}| < \infty$.

Then, if $T < T_1$, each of (i)_T, (ii)_T, (iii)_T implies each of (i)_T, (ii)_T, (iii)_T.

Proof. (i)_T $\Rightarrow$ (ii)_T. The heat equation remains valid for the derivatives of g , and therefore

$$\partial_t^n (z^{2n} g) = \partial_z^{2n+1} g = \partial_t^n (z \partial_z^{2n} g).$$

Since $g(0, t)$ is holomorphic for $|t| < T_1$, if $T < T_2 < T_1$, then at $(0, 0)$

$$|\partial_t^n g / n!| < O(1/T_2^n),$$

hence

$$|\partial_z^{2n} g| = O(n! / T_2^n),$$

and an analogous estimate holds for $\partial_z^{2n+1} g$. The Taylor series of f then gives

$$|f(z)| < \sum_n \binom{2n}{n} \frac{|z|^{2n}}{(2n)!} \frac{n!}{T_2^n}.$$

One has

$$\frac{1}{2n+1} \binom{2n}{n} \leq \frac{2^{2n}}{(n!)^2},$$

so

$$|f(z)| < O \left(\sum_n \frac{1}{n!} \left(\frac{|z|^2}{4T} \right)^n \right),$$

and this sum is $\exp(|z|^2/4T)$.

(ii)_T $\Rightarrow$ (i)_T. It suffices to take g given by the heat kernel:

$$g(z, t) = \frac{1}{\sqrt{4\pi t}} \int_{i\mathbb{R}} f(z + x) \exp(-x^2/4t) dx.$$

Changing x to $-ix$ changes the orientation of the cycle $i\mathbb{R}$ and the sign of the integrand, and hence does not change the integral. Condition (ii) ensures convergence for $|t| < T$.

The relation between (ii) and (iii) is clear: (ii)_T $\Rightarrow$ (iii)_T, and (iii)_T $\Rightarrow$ (ii)_T is verified by writing $f(z)$ as the mean of its values on a disk centered at z and of radius $1/|z|$, and by estimating that mean by Cauchy-Schwarz.

6

In homage to the Stone–von Neumann theorem, the bundle $\mathscr{W}$ on S is, projectively, endowed with a holomorphic connection for which the action of H is horizontal.

Let us make precise what “projectively” means.

On the projective line S there is the ample invertible sheaf $\mathcal{O}(1)$. Let $\mathcal{O}(-1)$ be its dual. A square root $\mathcal{O}(-\frac{1}{2})$ of $\mathcal{O}(-1)$ is an invertible sheaf L together with an isomorphism $L^{\otimes 2} \xrightarrow{\sim} \mathcal{O}(-1)$. Locally such a square root exists and, locally, two square roots are isomorphic, but the isomorphism is not unique: there is a ± 1 ambiguity. This ambiguity obstructs the global existence of $\mathcal{O}(-\frac{1}{2})$. It is not on $\mathscr{V}$, but on

$$\mathscr{V}(-\tfrac{1}{2}) := \mathscr{V} \otimes \mathcal{O}(-\tfrac{1}{2}) \quad (6.1)$$

that there is a connection

$$\nabla: \mathscr{V}(-\tfrac{1}{2}) \rightarrow \mathscr{V}(-\tfrac{1}{2}) \otimes \Omega^1.$$

Note that if $\mathcal{O}(-\frac{1}{2})'$ and $\mathcal{O}(-\frac{1}{2})''$ are two square roots of $\mathcal{O}(-1)$, and ∇ is a connection on $\mathscr{V} \otimes \mathcal{O}(-\frac{1}{2})'$, its transport to $\mathscr{V} \otimes \mathcal{O}(-\frac{1}{2})''$ by an isomorphism of square roots

$$\alpha: \mathcal{O}(-\tfrac{1}{2})' \xrightarrow{\sim} \mathcal{O}(-\tfrac{1}{2})''$$

does not depend on the choice of α , which can be changed locally only by a constant ± 1 . Thus the notion of a “connection on $\mathscr{V}(-\frac{1}{2})$ ” is well defined, independently of the global existence or choice of $\mathcal{O}(-\frac{1}{2})$.

Let $D \in S$ and let t be a tangent vector at D . For s a local section of $\mathscr{V}(-\frac{1}{2})$, we must define $\nabla_t s$ in the fiber of $\mathscr{V}(-\frac{1}{2})$ at D . We proceed as follows. After making an auxiliary choice u , we define $\nabla_t^u s$ for s a local section of $\mathscr{V}$, or of $\mathcal{O}(-\frac{1}{2})$. This derivation ∇_t^u depends on u , with dependence of the form

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) + a \cdot s(D),$$

where a takes opposite values for $\mathscr{V}$ and for $\mathcal{O}(-\frac{1}{2})$. For $s = s_1 \otimes s_2$, with s_1 a local section of $\mathscr{V}$ and s_2 a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t(s) := \nabla_t^u(s_1)s_2(D) + s_1(D)\nabla_t^u(s_2)$$

is independent of the choice of u .

Let $D \in S$ be a line in $\mathbb{C}^2$. A tangent vector t at $D \in S$ is identified with a linear map $t: D \rightarrow \mathbb{C}^2/D$, i.e. with an element of $(\mathbb{C}^2/D) \otimes D^*$. To $u' \otimes v' \in (\mathbb{C}^2/D) \otimes D^*$ one attaches $t(d) = \langle u, d \rangle v'$, where u is any representative of u' . Our auxiliary datum will be a lift of t , i.e. a lift of t to $\mathbb{C}^2/D \otimes \mathbb{C}^2$. We write it as $t = u' \otimes u$, with $u \in \mathbb{C}^2$ mapping to u' in $\mathbb{C}^2/D$.

The tangent space at the origin of H is $\mathfrak{z} = \mathbb{C}^2 \times \mathbb{C}$. For z in this tangent space, let ∂_z be the corresponding right-invariant vector field. One has

$$[\partial_x, \partial_y] = -\langle x, y \rangle'$$

with, on the right, the bracket in the Lie algebra of H . For $x, y \in \mathbb{C}^2$, computing $[x, y]$ as an infinitesimal commutator, one obtains

$$[\partial_x, \partial_y] = -2\pi i \langle x, y \rangle (\partial_0, \partial_0, 1). \quad (6.2)$$

Let $\mathscr{W}_\bullet$ be the quotient of the enveloping algebra of the Lie algebra of right-invariant vector fields by the relation $\delta(0, 0, 1) = 1$. In $\mathscr{W}_\bullet$, one deduces from (6.2) that, for $d \in D$,

$$[\partial_u^2, \partial_d] = 2 \cdot 2\pi i \langle u, d \rangle \partial_u = \langle u, d \rangle \partial_u = \langle \bar{t}(d) \rangle. \quad (6.3)$$

Compute to first order around D . For $\varepsilon^2 = 0$ and $\varphi_0(g) \in W_D$, so that $\partial_d \varphi_0 = 0$, one has

$$\partial_{d+\varepsilon t(d)}[\varphi_0(g) + \varepsilon \partial_u^2 \varphi_0(g)/(4\pi i)] = \varepsilon([\partial_d, \partial_u^2] \varphi_0(g) + \partial_d \partial_u^2 \varphi_0(g)) = 0,$$

since $\partial_d(\partial_u^2 \varphi_0/(4\pi i)) = -[-\partial_u^2/4\pi i, \partial_d] \varphi_0$; apply (6.3). To first order around D , the function of g and D' defined by

$$\varphi(g, D + \varepsilon t) = \varphi_0(g) + \varepsilon(\partial_u^2/4\pi i) \varphi_0(g)$$

is therefore a section of the bundle $\mathscr{V}$. If $\varphi(g, D')$ is a local section of $\mathscr{V}$, this gives the derivative ∇_t^u in W_D :

$$\nabla_t^u = \partial_t \varphi((g, D)) + (\partial_u^2/4\pi i) \varphi(g, D). \quad (6.5)$$

Let us compute the dependence of (6.5) on u . If u is replaced by $u + x$ with $x \in D$, then in $\mathscr{W}_\bullet$,

$$\partial_{u+x}^2 = (\partial_u + \partial_x)^2 = \partial_u^2 + 2\partial_u \partial_x + [\partial_x, \partial_u] = \partial_u^2 + 2\partial_u \partial_x + 2\pi i \langle u, x \rangle. \quad (6.6)$$

For $\varphi \in W_D$, one has $\partial_x \varphi = 0$, and (6.6) simplifies to

$$\partial_{u+x}^2 \varphi = \partial_u^2 \varphi + 2\pi i \langle u, x \rangle \varphi, \quad (6.7)$$

hence

$$\partial_{u+x}^2 \varphi/(4\pi i) = \partial_u^2 \varphi/(4\pi i) + \frac{1}{2} \langle u, x \rangle \varphi. \quad (6.8)$$

The choice of u also allows one to define the derivative ∇_t^u with respect to t of a local section of $\mathcal{O}(1)$. A section of $\mathcal{O}(1)$ on $U \subset S$ is a homogeneous function of degree 1 on the inverse image of U in $\mathbb{C}^2$, and one differentiates with respect to $\langle u, d \rangle \partial_u$ to obtain a homogeneous function of degree 1 on D . When u is changed to $u + x$,

$$\nabla_t^{u+x}(s) = \langle u + x, d \rangle \partial_{u+x} s = \nabla_t^u(s) + \langle u, x \rangle s.$$

This derivation induces one on $\mathcal{O}(-\frac{1}{2})$, with

$$\nabla_t(s^{-1/2}) = -\frac{1}{2} \nabla_t(s) \cdot s^{-3/2},$$

and, for dependence on u , where now s is a local section of $\mathcal{O}(-\frac{1}{2})$,

$$\nabla_t^{u+x}(s) = \nabla_t^u(s) - \frac{1}{2} \langle u, x \rangle s. \quad (6.9)$$

The dependences on u in (6.8) and (6.9) cancel, giving a derivation ∇_t for $\mathscr{V}(-\frac{1}{2})$.

Since the connection ∇ is defined in terms of right-invariant vector fields on H , it commutes with the action of H .

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We now write the connection of no. 6 in coordinates. Choose coordinates as in no. 1 and let E be the x -axis. On S , $\mathcal{O}(1)$ has a section s with a simple zero at E . It is unique up to a factor and trivializes $\mathcal{O}(1)$ on $S - E$. Thus there is a square root $\mathcal{O}(-\frac{1}{2})$ on $S - E$, trivialized by a square root $s^{-1/2}$ of s^{-1} . By such a trivialization, the desired connection is transported to a connection on the bundle $\mathcal{V}$. We compute it.

Let D_t be the line generated by $(-t, 1)$. Via r_E in (2.1), a local section of $\mathcal{V}$ is identified with a function $f(z, t)$ holomorphic in z and t . We compute its derivative with respect to t . In no. 6, one may take for t the map

$$D_t \rightarrow \mathbb{C}^2, \quad (-tz, z) \mapsto (-1, 0)z;$$

this is $d \mapsto -\langle (1, 0), d \rangle (1, 0)$.

The corresponding derivation of $d^* \in \mathcal{O}(1)$ annihilates s , so one simply has

$$\nabla_t f = (\partial_t - \partial_z^2 / 4\pi i) f, \quad (7.1)$$

and the local horizontal sections are the solutions of the heat equation

$$(\partial_t - \partial_z^2 / 4\pi i) f = 0. \quad (7.2)$$

From no. 5 one deduces that through any $s \in W_D^?(-\frac{1}{2})$ there passes a local horizontal section of $\mathcal{V}(-\frac{1}{2})$. It is automatically a section of $\mathcal{V}^?(-\frac{1}{2})$. It exists in a neighborhood of D whose size increases as s grows more slowly. Notice that in infinite dimension a connection need not define a local trivialization, even if through every point there passes a local horizontal section.

The connection ∇ on $\mathcal{V}(-\frac{1}{2})$ defines one on the corresponding projective-space bundle

$$\mathbb{P}(\mathcal{V}) = \mathbb{P}(\mathcal{V}(-\frac{1}{2})).$$

Its local horizontal sections are the images of nonzero horizontal sections of $\mathcal{V}(-\frac{1}{2})$. Although the sphere S is simply connected, one has the following.

8. Surprise

The projective-space bundle $\mathbb{P}(\mathcal{V})$ over S has no global horizontal section.

The argument of no. 1 would already lead one to expect this: apply it to the projective representation of $\mathrm{SL}(2, \mathbb{C})$ given by the action of $\mathrm{SL}(2, \mathbb{C})$ on the projective space of global horizontal sections of $\mathbb{P}(\mathcal{V})$.

First proof. Divide S into two hemispheres S_1 and S_2 and choose $\mathcal{O}(-\frac{1}{2})$ on each S_i . On $S_1 \cap S_2$ one has, locally, an isomorphism

$$\alpha: \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2,$$

defined up to sign; but α can be defined globally only on the double cover of the circle $S_1 \cap S_2$, with monodromy -1 . If there were global horizontal projective sections of $\mathbb{P}(\mathcal{V})$, there would be horizontal sections f_1 and f_2 of $\mathcal{V}(-\frac{1}{2})$ on S_1 and S_2 which, on $S_1 \cap S_2$, would agree up to sign. On $S_1 \cap S_2$, this would allow us to normalize

$$\alpha: \mathcal{O}(-\frac{1}{2})_1 \xrightarrow{\sim} \mathcal{O}(-\frac{1}{2})_2$$

by requiring $\alpha(f_1) = f_2$, contradicting the nonexistence of α on $S_1 \cap S_2$.

Second proof. Choose coordinates as in no. 1. Let E be the x -axis and D the y -axis. As in no. 2, identify W_D with the space of entire functions $f(z)$. Let $f \neq 0$ in W_D . By the proposition, for there to exist on $S - E$ a horizontal section of $\mathbb{P}(\mathcal{V})$ passing through f , it is necessary and sufficient that f satisfy the growth condition

$$\forall \varepsilon > 0 \quad |f(z)| < O(\exp(\varepsilon|z|)). \quad (8.1)$$

If E' is another line, then by (4.1), the existence of a horizontal section on $S - E'$ passing through f is expressed by a growth condition

$$\forall \varepsilon > 0 \quad |f(z) \exp(az^2)| < O(\exp(\varepsilon|z|^2)), \quad (8.2)$$

with a suitable $a \neq 0$.

Put $g(z) = f(z) \exp(az^2/2)$. By (8.1) and (8.2), $|g(z)|$ is bounded by a multiple both of $|\exp(\varepsilon|z|^2 + az^2/2)|$ and of $|\exp(\varepsilon|z|^2 - az^2/2)|$. Thus, in every direction except those for which az^2 is purely imaginary, g has quadratic exponential decrease. In all directions it has, uniformly, at most quadratic exponential growth. By Phragmén–Lindelöf, g has quadratic exponential decrease in every direction; hence it is zero, and $f = 0$.

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In the coordinates of no. 1, let D_t be the line generated by $(-t, 1)$. Let $d = (-tz, z)$. For the commutator of d and $\bar{d}$ one has

$$(d, \bar{d}) = \exp(-2\pi i(t - \bar{t})|z|^2) \in \mathbb{C}^* :$$

the commutator is real positive, and

$$(d, \bar{d}) > 1 \iff \text{Im}(t) > 0.$$

The lines D for which $(d, \bar{d}) > 1$ for a generator d of D therefore form a disk in S , said to be attached to the real structure $H_{\mathbb{R}}$.

The real structures $L \subset \mathbb{C}^2$ on which $\langle \cdot, \cdot \rangle$ is real form a homogeneous space under $\text{SL}(2, \mathbb{C})$, with $\text{SL}(2, \mathbb{R})$ the stabilizer of $\mathbb{R}^2$. Each defines a real structure H_L on H , hence a complex conjugation σ_L with fixed points $L \cdot \text{U}^1$.

The disks in S likewise form a homogeneous space under $\text{SL}(2, \mathbb{C})$, the stabilizer of the disk associated with $H_{\mathbb{R}}$ being $\text{SL}(2, \mathbb{R})$. The correspondence

$$L \mapsto X(L) := \{D \mid \langle d, \sigma_L(d) \rangle > 1 \text{ for } d \text{ a generator of } D\}$$

is therefore a bijection from the space of the real structures under consideration to the space of disks in S .

Passing from a disk to the complementary disk corresponds, on real structures, to $L \mapsto iL$.

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The computations of no. 3 admit the following interpretation. Let X be an open disk in S , corresponding to a real structure L , hence to real forms H_L of H and $\mathrm{SL}(L)$ of $\mathrm{SL}(2, \mathbb{C})$. Then $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$ has an H_L -invariant pre-Hilbert structure, and its completion

$$\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})^\wedge$$

is the irreducible unitary representation of H_L . One has

$$\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0}) \subset \Gamma(X, \mathcal{V}|_{V=0}) \subset \Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})^\wedge.$$

The double cover $\mathrm{SL}(L)'$ of $\mathrm{SL}(L)$ acts on $\mathcal{V}(-\frac{1}{2})|_X$, compatibly with its action on $\mathcal{O}(-\frac{1}{2})$, and therefore acts on $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$. This is the action which normalizes the action of H_L and induces the natural action of $\mathrm{SL}(L)'$ on H_L .

The subrepresentation $\Gamma(X, \mathcal{V}(-\frac{1}{2})|_{V=0})$ may be interpreted as the space of real-analytic vectors for the action of $\mathrm{SL}(L)'$.

This description shows that the projective representation of $\mathrm{SL}(L)$ extends to the monoid of those $g \in \mathrm{SL}(2, \mathbb{C})$ such that $gX \supset X$.

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Everything above generalizes from $\mathrm{SL}(2, \mathbb{C})$ to $\mathrm{Sp}(2n, \mathbb{C})$: replace $\mathbb{C}^2$ by $\mathbb{C}^{2n}$, endowed with a nondegenerate alternating form $\langle \cdot, \cdot \rangle$ that is real on $\mathbb{R}^{2n}$; replace H by $\mathbb{C}^{2n} \times \mathbb{C}^*$ with group law

$$(x, \lambda)(y, \mu) = (x + y, \lambda\mu \exp(\pi i \langle x, y \rangle));$$

and replace S by the space of Lagrangian, i.e. maximal isotropic, subspaces D of $\mathbb{C}^{2n}$. The disk attached to a real structure becomes the Siegel space of those D for which $|\langle d, \bar{d} \rangle| > 1$ for $d \neq 0$ in D . The invertible sheaf $\mathcal{O}(-\frac{1}{2})$ is to be replaced by $\det(D)^{-1/2}$.

12. Remark

Alongside representations of H in which $U^1 \subset \mathbb{C}^*$ acts by multiplication by λ , consider similarly representations in which the action is by λ^{-1} . One defines, as in no. 2, a holomorphic bundle $\mathcal{V}'$ of such representations over S ; on $\mathcal{V}'(-\frac{1}{2})$ there is again a connection, given by a heat equation.

On a connected and simply connected open set U of S , consider $\mathcal{W} = \mathcal{V}(-\frac{1}{2})$, its connection, and the action of H . In $\Gamma(U, \mathcal{W})|_{V=0}$, for each $x \in U$ one has a linear form fixed by the corresponding line, defined up to a factor: $f \mapsto f(\xi)$ for $f \in \mathcal{V}$. For each $x \in U$, there exists a vector $v(x)$ fixed by D_x ; in the Schrödinger model, when x corresponds to $E =$ the x -axis, this is the constant 1. It is unique up to a factor.

Fix the real structure $H_{\mathbb{R}}$, hence a real structure $S_{\mathbb{R}} \subset S$, a circle, and a decomposition of $S - S_{\mathbb{R}}$ into two disks U_1 and U_2 . Then:

- (a) The complex conjugate of $\Gamma(U_1, \mathcal{W})|_{V=0}$ is $\Gamma(U_2, \mathcal{W}')|_{V=0}$.
- (b) The Hermitian form on $\Gamma(U_1, \mathcal{W})|_{V=0}$, invariant under $H_{\mathbb{R}}$, can be reinterpreted as a bilinear pairing between $\Gamma(U_1, \mathcal{W})|_{V=0}$ and $\Gamma(U_2, \mathcal{W}')|_{V=0}$ that is invariant under H .

Question. Let C be a Jordan curve in S , separating $S - C$ into two open sets U_1 and U_2 . Is there a unique separating H -invariant bilinear form pairing $\Gamma(U_1, \mathscr{W})|_{V=0}$ and $\Gamma(U_2, \mathscr{W}')|_{V=0}$, induced by a perfect duality between $\Gamma(U_1, \mathscr{W})|_{V=0}$ and $\Gamma(U_2, \mathscr{W}')|_{V=0}$?

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